A Visual Analytics Solution for Analyzing and Mining Infectious Disease Data

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***Abstract*—In this paper, we present a visual analytics solution to analyze and visualize infectious disease data (e.g., COVID-19 data). During the peak of COVID-19 pandemic we observed large jumps in infections, we delve into the link between the COVID-19 related infection rates and population density. We visually discover knowledge like relationships between population density and infection rates in health regions, as well as connections between infection rates and socio-economic characteristics. Evaluation on real-life Canadian COVID-19 data show the practicality of our visual analytics solution—which employs visualization, data mining techniques and statistical analysis—in providing valuable insights into the dynamics of pandemics.**

Keywords—information visualization, artificial intelligence (AI), visual knowledge discovery, health informatics, data mining, COVID-19, hospitalization, infection, population, health region

I. Introduction and Related Works

In the current era of big data, data are everywhere. Embedded in these data are implicit, previously unknown and useful knowledge that can be discovered by big data science [1]—which makes good use of data mining [2-7], machine learning [8], statistics [9], and visualization [10, 11]. The latter has been applied to many real-world application areas like transportation [12], music [13], biomedicine [14-17], social networks [18, 19], uncertain data [20], and time series [21, 22]. This paper focuses on health informatics.

The coronavirus disease 2019 (COVID-19) pandemic emerged in late 2019 and rapidly developed into a global health crisis. Using data mining technologies, we set out to uncover some of these unexplained patterns, such as the relationship between infection rates and hospitalizations, the connection between population density and infection rates, as well as the link between infection rates and socio-economic characteristics. More importantly, using data mining techniques on the COVID-19 in Canada dataset provides an effective and efficient method for revealing non-trivial patterns, and correlations. We focus on the analysis of a large amount of COVID-19 data from Canada, such as infection rates, hospitalizations, cumulative and daily cases in different provinces and health regions.

In our study, we specifically explored data at the health region level. We examined factors such as infection rates, population density, and peer groups categorized by principal characteristics. Health regions, being smaller and more numerous than provinces, offer more localized data. This approach enables a deeper understanding of how local social and economic conditions in specific areas influence the spread of the virus.

Moreover, we place significant emphasis on data visualization in our paper. Given the extensive amount of data generated in the healthcare sector, especially concerning epidemic diseases like COVID-19, employing visual representations is crucial. These visuals simplify complex data, making it more accessible and understandable, which is vital in aiding the fight against the disease.

There have been tools [23] for visualizing and analyzing COVID-19 data that has been presented, emphasizing the importance of data visualization in understanding complex epidemiological data and its potential applications in other domains. Various interactive dashboards (e.g., those by Johns Hopkins University and the *New York Times*) [24] provide real-time insights into the pandemic's growth through formats (e.g., bubble maps, choropleth maps, heatmaps). Time-series alignment for comparison, a feature that aligns time-series data by specific thresholds to compare how different regions have passed through specific stages of the pandemic. There is a study [25] to visualize the global spread of the COVID-19 pandemic over the first 90 days, through the principal component analysis approach of dimensionality reduction, correlation matrices, and standardized plots.

Application of advanced data analytics and machine learning techniques to predict outbreaks, optimize resource allocation, or improve diagnostic tools has been presented [26]. There has been analysis of Google Trends data to understand public sentiment regarding lockdown measures, vaccine rollouts, and government responses to the pandemic [27]. Supervised machine learning models [28] (e.g., logistic regression, artificial neural network) have been used for diagnosing COVID-19. The findings indicate these models could significantly aid in COVID-19 diagnosis, offering potential relief to strained healthcare systems.

In Canada, each province and territory are responsible for its own healthcare management. Hence, COVID-19 data are collected by various provincial health authorities, leading to a diverse and decentralized dataset. Within each province (e.g., Manitoba), data relevant to its specific region are collected by different Regional Health Authorities (RHAs) (e.g., Winnipeg RHA), which include data like infection rates, hospitalization data, testing results, and vaccination rates.

A spatial data science system [29] for analyzing COVID-19 data emphasizes the importance of understanding the geographical aspects of the disease's spread and its implications. The focus is on spatial data analytics across different geographic locations and the spatial hierarchy offers a more efficient,

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insightful, and user-friendly way to understand and conduct spatial data analytics at higher granularity.

Key contributions of this paper include (a) our design of a visual analytics solution to visualize infectious disease distribution and to discover interesting patterns, as well as (b) its application to real-life COVID-19 data to demonstrate the practicality of our solution.

II. Our Visual Analytics Solution

Let us demonstrate our visual analytics solution by using real-world Open COVID-19 API, which provides a definitive source for data regarding the COVID-19 pandemic in Canada. There are three groups of datasets [30]: (a) health region-level (hr), for case and death data only; (b) province/territory-level (pt), for all data types; (c) country-level (can), for all data types.

This enables statistical analysis on all provinces in Canada for various factors contributing to COVID-19. We design and develop our visual analytics solution to visually discover knowledge from variables like daily deaths, infections, and vaccinations. We also utilize publicly available data (e.g., Government of Canada statistics "Population and dwelling counts: Canada, provinces and territories" and "Health Regions: Boundaries and Correspondence with Census Geography"). These data capture each province's or health region's population and its population density. The data provide an important background for our analysis, and it comes from official government sources, representing authority and accuracy. This is an invaluable resource for our study.

A. Overall Trend of the Disease in a Country

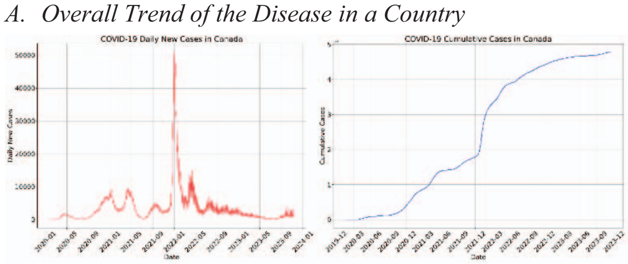

Fig. 1. (a) New daily COVID-19 cases and (b) cumulative cases in Canada

Fig.1(a) is a representation of the daily new cases of COVID-19 in Canada over a certain period. The graph exhibits several waves of infection, which is characteristic of the COVID-19 pandemic. These waves correspond to periods where new daily cases rise sharply and then fall. The figure shows the first few waves show a gradual increase in amplitude, with each subsequent peak being higher than the last. This suggests that either the virus was spreading more extensively, or testing was becoming more widely available and capturing more cases. The most significant peak in January 2022 appears to be an outlier, with a very sharp spike that far exceeds any other point on the graph. This was due to the emergence of a new and more infectious variant Omicron, a super-spreader event, or it could be a data anomaly such as a backlog of cases being reported in a single day due to delayed testing results or reporting. After the largest peak, the number of daily new cases drops significantly but remains volatile, with several smaller spikes, indicating ongoing transmission and outbreaks. The "noisy" or jagged appearance of the line, especially in the latter part of the graph, suggests day-to-day variability in reported case numbers. This could be influenced by factors such as testing availability, reporting delays over weekends and holidays, or fluctuations in transmission. Similarly, line chart in Fig. 1(b) shows cumulative cases.

B. Overall Trend of Cases across Province

Fig. 2 shows cumulative and daily new COVID-19 cases by province in Canada. They clearly show that Ontario and Quebec, being the provinces with the highest populations, naturally report the greatest number of COVID-19 cases. Furthermore, we explore the correlation between population density and infections. For example, we observed plateau at a similar cumulative infection rate by population in provinces of Alberta and Nova Scotia by population percentage, except Nova Scotia's curve is a far sharper spike. This can be tied to a higher population density. When we look closer at hospitalizations and deaths between these two provinces, there are further differences. As Nova Scotia's infections were at a steeper curve, so were the deaths. Moreover, we observed that the deaths plateau at a lower percentage than Alberta. Similarly, Nova Scotia consistently had fewer hospitalizations and ICU cases by population percentage than Alberta.

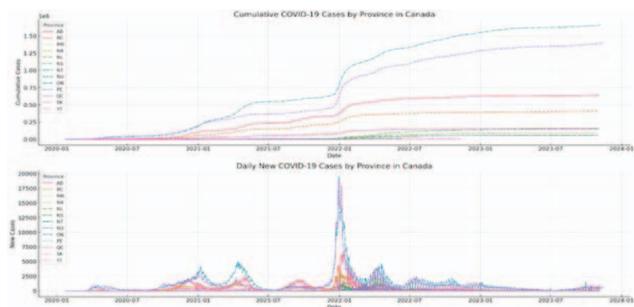

Fig. 2. COVID-19 cumulative and daily new cases by province.

C. Cases and Hospitalizations

Finding a sequential mathematical relationship between daily new COVID-19 cases and hospitalizations is a complex task, as the relationship can be influenced by numerous factors. However, a common approach to explore such relationships is through correlation and time series analysis. For instance, the Pearson correlation analysis involves calculating the correlation coefficient between daily cases and hospitalizations. A strong correlation might indicate a direct relationship. The Pearson correlation coefficient [31] between daily new COVID-19 cases and daily hospitalizations in Canada is approximately 0.37. This indicates a moderate positive correlation. In other words, there is some degree of linear relationship between the two, but it is not very strong. Often, hospitalizations lag behind cases because there is a delay between when people get infected, when they develop symptoms severe enough to require hospitalization, and when these hospitalizations are reported. We analyze this by introducing a lag in the cases data and then calculating the correlation. The typical time lag could range from a few days to a couple of weeks. The strongest correlation between daily new COVID-19 cases and hospitalizations in Canada occurs with a 4-day lag as shown is Fig. 3(a), and the Pearson correlation coefficient at this lag is approximately 0.41. This suggests a slightly stronger relationship than the immediate day-to-day correlation, indicating that hospitalizations tend to follow the trend in cases with a delay of about 4 days. It is important to note that correlation does not imply causation and that other factors might influence these trends.

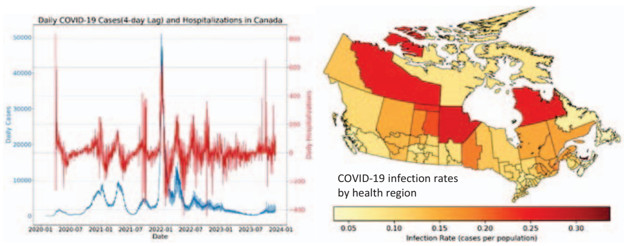

Fig. 3. (a) COVID-19 daily cases with 4-day lag and hospitalizations in Canada; (b) Infection rates by health regions in Canada

D. *Infection Rates and Population Density across Health Regions*

In our study, we use health regions instead of provinces to analyze COVID-19 data. Health regions are administrative areas defined by provincial or territorial governments in Canada. They are used for organizing and delivering health services, public health reporting, and health resource allocation. They are the smallest area unit we can get for COVID-19 infection data. According to the data in 2018, Canada had 100 health regions.

In recent years, there has been an increasing demand for relevant health information at a "community" level. As a result, health regions have become an important geographic unit by which health and health-related data are produced. Being smaller and more numerous than provinces, they provide more localized data. This allows for a better understanding of how a disease is spreading in specific areas, which can be crucial for targeted public health interventions. It visually distinguishes between urbanized areas and rural or wilderness areas, with a clear majority of Canadians living in urban areas.

Fig. 3(b) provides a visual representation of infection rates across health regions across Canada. There are clear variations in infection rates between different regions. Some regions show significantly higher rates, indicating more widespread infection relative to their population sizes. These differences could be influenced by numerous factors including population density, local public health policies, and access to healthcare services.

E. *Comparison between Health Regions within Provinces*

It is interesting (though may be counterintuitive) that many rural areas have higher infection rates than urban areas, as illustrated by the map. Take Manitoba as an example. The Northern Health Region exhibits the highest infection rate, significantly surpassing that of Winnipeg. This pattern also applies to other provinces (e.g., BC, Ontario).

In nearly every Canadian province, a frequent pattern emerges. Major population centers (e.g., Vancouver, Montreal, Toronto, Edmonton) have maintained moderate or even relatively low infection rates compared to less populated areas. On the other hand, regions like the Northern Health Region, Région du Nunavik, North Zone, and other remote areas, despite their low population density, have experienced significantly higher infection rates. Factors contribute to this phenomenon may include: (a) majority of Canada's rural population resides in small towns, where the density can be relatively high; (b) some rural areas experience colder temperatures year-round, creating conditions more conducive to the spread of viruses; (c) regions with robust healthcare infrastructure might have been more effective in testing and controlling the spread of the virus, potentially resulting in lower infection rates. This observation clearly indicates that population density alone does not fully explain the variations in infection rates.

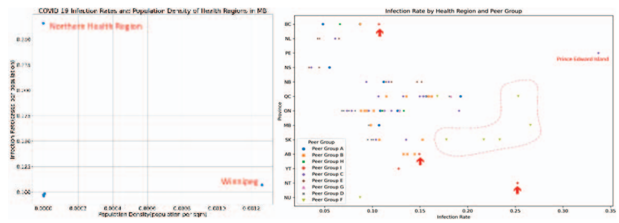

Fig. 4. (a) Infection rates and population density by health region in Manitoba; (b) Infection rates by health region and peer group

F. *Health Region Peer Groups*

To effectively compare health regions with similar socio–economic characteristics, health regions have been grouped into "peer groups". Statistics Canada used a statistical method to achieve maximum statistical differentiation between health regions. 24 variables were chosen to cover as many of the social and economic determinants of health as possible, using data collected at the health region level mostly from the Census of Canada. Concepts covered include (a) basic demographics (e.g., population change and demographic structure) and (b) living conditions (e.g., socio-economic characteristics, housing, and income inequality).

Peer groups based on 2017 health region boundaries and 2011 Census of Population, and 2011 National Household Survey data are available. There are currently nine peer groups identified by the letters A through I. There have been no changes made to peer group assignments since 2014 [32].

Fig. 4(b) is the graph representations of infection rates by health region and the peer group it belongs to. Each point on the graph represents a health region. The x-axis shows the infection rate for each health region. The y-axis corresponds to the province that the hr is in. Different colors and markers indicate the peer groups. This visualization makes it clearer to compare the infection rates across different health regions in provinces and observe any patterns or clusters within peer groups. Peer Groups A, B and C usually with high employment rates and urban areas has moderate or low infection rates. On the other hand, Peer Groups F and I have high or relatively high infection rates within their provinces:

- Peer Group F: Northern and remote regions, with very low percentage of visible minority population, and very high Indigenous population
- Peer Group I: Mainly rural and remote regions in the Western provinces and the Territories, with an average percentage of visible minority population, high

percentage of Indigenous population, and high employment rate.

These observations raise some important points about the dynamics of infectious disease spread in different settings. For instance, some high population density regions do not have very high infection rates, which suggests that high population density does not automatically translate to high infection rates. It could be indicative of effective public health measures in densely populated areas, such as more rigorous testing, tracing, and isolation protocols. It might also reflect better healthcare infrastructure and resources in more densely populated areas, which can help manage and control outbreaks more effectively.

Contrarily, Peer Groups F and I have higher COVID-19 infection rates. This pattern could be influenced by factors like healthcare resources, living conditions, comorbidities, public health infrastructure, employment, mobility and travel, cultural practices, information and education, resource allocation, as well as socioeconomic factors. These patterns underline the complexity of infectious disease dynamics. Factors like public health policy, community compliance with health guidelines, healthcare infrastructure, and even socio-economic factors play significant roles in determining how an infectious disease spreads within a community. This complexity is why tailored approaches are often necessary for different regions, considering their unique circumstances and challenges.

III. Conclusions

In this paper, we presented a visual analytics solution to analyze and visualize infectious disease data (e.g., COVID-19 data). Our solutions reveals interesting patterns like (a) general trends in a country, (b) common trends among the provinces/states, (c) their relationships with cases, hospitalization, population density, vaccination rates, as well as (d) their distributions among different health regions. Although we applied our solution to Canadian data, our solution can be easily applicable and adaptable to other geographic units (e.g., different continents, countries, provinces/states, health regions/jurisdictions). As *ongoing and future work*, we explore ways to further enhance our solution.

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